

Townsend's 2026 Monthly Dust Loss Model

Purpose

This model is used to estimate the percentage of potential monthly photovoltaic (PV) energy sacrificed due to dust soiling.

PV financing decisions are based on long-term energy estimates, and popular PV modeling software such as PVsyst include monthly inputs for soiling losses – snow in the winter, dust in the summer. This modeling tool relies principally on monthly rainfall to determine a changing pattern of soiling throughout the year. That pattern is also influenced by four other factors:

1. The daily soiling ramp rate (seasonally adjustable).
2. The number of manual array washings, up to two per year.
3. The influence of snow, where any month with at least a 5% projected snow loss will govern. This level of snow loss will treat that month's dust soiling loss as zero.¹
4. For bifacial arrays, the back side's contribution is presumed to have no appreciable dust buildup. The full array's monthly dust loss is reduced a little by the percentage of energy contributed from the always-clean back side.

Rules

The following table lists the core assumptions applied to each month, subject to the superseding influence of snow, which sets the dust loss to 0% if that month's snow losses are at least 5%. For months with snow less than 5%, the total loss = snow + dust – (snow x dust).

Monthly precipitation, inches (mm)	Soiling
≥4 (102)	0%
≥2 and <4 (51-102)	1%
≥0.5 and <2 (13-51)	If prior month precip. ≥2", add 8 days x seasonal ramp rate to prior month's soiling. If prior month precip. <2", add 15 days x seasonal ramp rate to prior month's soiling.
<0.5 (13)	Add product of number of days per month x seasonal ramp rate applicable to current month to prior month's soiling.

¹ Townsend's snow loss measurements during 2013-15 for seven orientations at Golden, CO yielded six months with one snow event. Over these 42 qualifying data points, the average monthly snow loss was 4.8%, so for this tool, we have rounded the mixed dust and snow threshold to 5%. This is viewed as the typical snow loss for which one 1" or greater snow event per month has occurred. One snow event of at least 1" is viewed as sufficient to clean the dust off. The average inches of snow for the one-event months was 4 inches; again, this is viewed as more than a sufficient amount of snow to clean off any dust.

A few other rules apply. 1) A maximum monthly loss of 30% is assumed. 2) A starting month must be chosen. We need a consistent, wrapped pattern of monthly dust losses applicable to all years. To do this, the month with the most precipitation is chosen as the starting month. The soiling applicable to that month is subject to the same rules described above, with one exception: if the starting month's precipitation is less than 2 inches, the soiling for that month is set to 2%. 3) A manual wash is assumed to occur on the first day of a month. Unlike the Kimber et al. modelⁱ, this model does not include a "grace period" of 0% soiling after a manual wash. The dust loss in a manual wash month will still be calculated based on a half month of accumulation, at the daily ramp rate applicable to that season.

If one or two manual washings per year are selected, the model does an exhaustive search, testing each month in sequence as the date of the first potential wash, then recomputing the *annual* energy loss associated with each potential choice. After checking each month, the wash month with the lowest annual soiling loss is deemed to be optimal. The same logic is used to determine when best to do a second wash. It is possible that the second-best time to do a manual wash may occur chronologically in a month earlier than the best first wash. **The incremental annual loss reduction value of the "second" wash is predicated on committing to also performing the more impactful "first" wash in its prescribed month.** Here, the labeling of "best" and "second-best" is probably more accurate than saying "first" or "second." If wash #1 is prescribed for August and wash #2 is prescribed for July, the low annual soiling loss being sought for two manual washes will only be attained if both the July and August washings occur.

Inputs

1. Twelve monthly precipitation amounts, specified in units of inches or millimeters.
2. Four seasonal daily dust accumulation (ramp) rates. Four rate choices are available per season, and the model has been automated to offer recommendations based on a combination of the annual rainfall and a qualitative determination of four local adder effects that increase soiling. The four rate choices bracket a wide and realistic range of observed daily dust accumulation rates:
 - a. 0.025%: mostly applies to ultra-sandy, inorganic particulates, <10"/yr rain
 - b. 0.05%: scrub desert, 10-15"/yr rain
 - c. 0.1%: typical default, safely applicable in many climates with >15"/yr rain
 - d. 0.15%, typical of summer/fall organic dust
3. Soiling adders. The above rates may be increased if certain conditions exist. Four qualitative choices can each add a +1 increment to the otherwise applicable rate, though the maximum accumulation rate is nevertheless capped at 0.15%/day. The four possible adders include:

- a. Agricultural, especially irrigated. This adder can also be checked if a site is affected by heavy pollen or stubborn surface contaminants such as musts or lichens, as have been observed in California's wine country or in humid southeastern states.
 - b. Under a commercial airport flight path and within 5 km of a runway. For guidance, a flight path is assumed to be 2 km wide, so this condition would apply to any 10 km² rectangle on either end of a runway.
 - c. Prone to heavy bird droppings. Practically, this implies the array is close to oceans, lakes, and rivers where large flocks are known to congregate.
 - d. Prone to urban automotive and industrial emissions. For simplicity, this condition is advised within cities of at least 250,000 people.
4. Number of manual washes, from 0 to 2 per year.
 5. Bifacial: If bifacial, a separate calculation linked to the snow model is used to determine the monthly percentage of generation attributable to the essentially unsoiled back side, with the overall dust loss reduced accordingly.

Ramp Rates

The following table shows the recommended seasonal daily ramp rates as a function of annual rainfall in the absence of any local soiling adder effects. These preliminary rates are then adjusted upward by the number of adders applicable to a given site.

Annual rainfall	Wettest quarter	2 nd wettest qtr.	2 nd driest qtr.	Driest quarter
<10"	0.025%	0.025%	0.025%	0.025%
10-15"	0.025%	0.025%	0.025%	0.05%
15-25"	0.025%	0.05%	0.1%	0.15%
25-40"	0.05%	0.05%	0.1%	0.1%
>40"	0.1%	0.1%	0.1%	0.1%

The following table amends the above rates by the number of rate increments based on the number of adders that apply. The final rate is clamped at a maximum of 0.15%/day, regardless of how many adders apply.

Quarter	0 adders	1 adder	2 adders	3 adders	4 adders
Wettest	No change	No change	No change	+1	+2
2 nd wettest	No change	No change	+1	+2	+2
2 nd driest	No change	+1	+1	+2	+3
Driest	No change	+1	+2	+3	+3

As an example, consider a location such as Oakland, California, with an average annual rainfall of 24 inches. The urban adder would apply based on the city's population of 440,000 since this exceeds the nominal 250,000 population threshold. Depending on where in the city a project was being considered, the bird presence and airport flight path adders could plausibly apply, but for this example, we'll assume only one soiling adder will apply. Per the second table above, the "1 adder" column adds one rate increment to the two driest quarters. This increases the fall rate from 0.1%/day to 0.15%/day. The summer rate was already at its maximum allowable value, so it remains unchanged.

Oakland rainfall	Wettest (winter, 13.7")	2 nd wettest (spring, 5.6")	2 nd driest (fall, 4.5")	Driest (summer, 0.2")
15-25"	0.025%	0.05%	0.15%	0.15%

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ⁱ Kimber, A., Mitchell, L., Nogradi, S. and Wenger, H. (2006) The Effect of Soiling on Large Grid-Connected Photovoltaic Systems in California and the Southwest Region of the United States. Proceedings of the 4th IEEE World Conference on Photovoltaic Energy Conference, Waikoloa, 2391-2395.
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